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Generalized continuous nowhere differentiable function families

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The existence of continuous nowhere-differentiable functions is a fundamental phenomenon in real analysis, challenging the classical intuition that continuity generally implies differentiability. Once regarded as pathological constructions, such functions are now recognized as abundant in the space of continuous functions, both in the sense of Baire category and through notions of prevalence. In this paper, we develop a general framework for constructing continuous nowhere differentiable functions represented by trigonometric series with decaying amplitudes and rapidly increasing frequencies. We establish a sufficient condition for nowhere differentiability that applies to a broad class of Fourier-type series, thereby providing a unified criterion that encompasses several classical constructions, including the Weierstrass function and related examples. The proposed approach also yields new families of continuous nowhere differentiable functions under flexible conditions on the amplitudes and frequencies. These results contribute to a systematic understanding of the mechanisms underlying the breakdown of differentiability and further illustrate the richness and ubiquity of nowhere-differentiable behavior within the class of continuous functions.

Keywords: Continuous functions; non-differentiability; Fourier series; harmonic analysis; pathological functions; Baire category; prevalence; fractals; Brownian motion.

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1. Introduction

The emergence of continuous nowhere differentiable functions profoundly transformed the nineteenth-century understanding of continuity and regularity. At a time when continuity was often associated with a certain degree of local smoothness, it was natural to expect that a continuous function on $\mathbb{R}$ would be differentiable at most points, with non-differentiability occurring only at a small number of exceptional locations. This intuition is reflected, for example, in the work of Ampère, who argued as early as 1806 that continuity should be closely related to differentiability.[1] Such a viewpoint was consistent with the prevailing development and applications of differential calculus and remained influential until the emergence of explicit counterexamples. Among the earliest known constructions is the example attributed to Bolzano in the 1830s. Although his work remained unpublished until the beginning of the twentieth century,[2,3] it contains a construction of a continuous function that fails to possess a finite derivative at every point, now commonly referred to as the *Bolzano function*. This example provided an early and striking demonstration that continuity alone does not guarantee differentiability at any point. Later, in 1861, Riemann introduced a continuous trigonometric series whose differentiability properties were markedly more irregular, providing an important step toward the systematic study of continuous functions with pathological differentiability behavior.[4] These developments progressively challenged the prevailing intuition and paved the way for the systematic investigation of nowhere differentiable functions, culminating in the celebrated construction of Weierstrass and the subsequent development of a rich theory of irregular continuous functions. A major conceptual turning point came with the publication of Weierstrass's celebrated example in 1872.[5] Represented by a trigonometric series with rapidly increasing frequencies and appropriately decaying amplitudes, this construction provided an explicit continuous function that is nowhere differentiable. It demonstrated decisively that continuity and differentiability are fundamentally distinct notions of regularity. In 1916, Hardy established a sharp refinement of the differentiability analysis for this class of Weierstrass-type functions, showing that the relevant parameter conditions cannot, in general, be substantially weakened.[5] These developments initiated a systematic study of highly irregular continuous functions and substantially changed the understanding of the relationship between continuity and smoothness. Subsequent research revealed that nowhere differentiable functions are by no means isolated pathological examples. In the space $C([0,1])$ of continuous real-valued functions equipped with the uniform norm, the set of nowhere differentiable functions is topologically large: in particular, it contains a dense G_δ subset, a phenomenon closely connected with the development of the Baire category method.[6] From a different perspective, prevalence theory provides a measure-theoretic analog of genericity and shows that nowhere-differentiable behavior can also be abundant in an appropriate infinite-dimensional sense.[7] Thus, what initially appeared to be an exceptional pathology is now understood as a structurally rich and widespread phenomenon

within spaces of continuous functions. The emergence of such examples nevertheless challenged deeply rooted mathematical intuition. Their existence was initially met with considerable skepticism and discomfort, as reflected in contemporary reactions to the increasingly irregular functions constructed during the nineteenth century.[8,9] Rather than remaining isolated counterexamples, however, nowhere differentiable functions have become fundamental objects in modern analysis and probability. A particularly striking manifestation occurs in stochastic processes: almost every sample path of standard Brownian motion is continuous but nowhere differentiable.[10,11] This connection between deterministic constructions and stochastic phenomena demonstrates that nowhere differentiability is not merely a pathological feature of artificial examples but an intrinsic form of irregularity that arises naturally in mathematical models. Beyond their role as counterexamples to classical intuitions about regularity, nowhere-differentiable functions also provide fundamental prototypes for fractal curves, whose intricate geometric structures have become a central subject of twentieth-century mathematics. Building on this historical and analytical perspective, we develop a systematic construction framework in this article, grounded in harmonic analysis. In particular, we investigate trigonometric series of the form:

$$f(x) = \sum_{j \geq 0} \alpha_j e^{i\xi_j x},$$

where the coefficients (α_j) tend to zero, whereas the associated frequencies (ξ_j) exhibit sufficiently rapid growth. We first derive general conditions ensuring the convergence and continuity of the resulting trigonometric series. We subsequently establish a sufficient condition for nowhere differentiability, formulated in terms of the interaction between the decay of the coefficients and the growth of the frequencies. This criterion encompasses and unifies several classical constructions, including those of Riemann, Weierstrass, and Hardy, and provides a flexible framework for generating new classes of continuous nowhere-differentiable functions that exhibit rich, potentially fractal geometric behavior. Our approach is therefore designed to elucidate the analytical mechanisms responsible for the complete breakdown of differentiability, while providing a unified framework for understanding a fundamental and far-reaching phenomenon in modern analysis. Hermite's famous reaction to the emergence of such pathological examples vividly illustrates the extent to which nowhere differentiable functions challenged the mathematical intuition of his time:

> "I turn away with dread and horror from this lamentable plague of continuous functions that have no derivatives."[8]

What was once regarded as a pathological and exceptional behavior is now known to be a widespread and structurally rich phenomenon, with nowhere differentiable functions forming a large class under several rigorous notions of genericity.

2. Construction of Nowhere Differentiable Continuous Functions

2.1. *Principal Theorem*

Theorem 1 (A Necessary Condition for Differentiability of Certain Complex Trigonometric Series). *Let $(\alpha_n)_{n\in\mathbb{N}}$ be a complex sequence, $(\xi_n)_{n\in\mathbb{N}}$ a sequence of strictly positive real numbers, and $\eta > 1$. Assume that the series $\sum \alpha_n$ converges absolutely and that $\xi_{n+1} \geq \eta\,\xi_n$ for all $n \in \mathbb{N}$. Finally, define for all $x \in \mathbb{R}$:*

$$f(x) = \sum_{j=0}^{+\infty} \alpha_j e^{i\xi_j x}.$$

The function f is continuous on $\mathbb{R}$. If it is differentiable at least one point, then

$$\lim_{n\to+\infty} \xi_n|\alpha_n| = 0.$$

Corollary 1 (Contrapositive form: A sufficient condition for nowhere differentiability). *Under the same hypotheses on (α_n) and (ξ_n), if*

$$\xi_n|\alpha_n| \not\longrightarrow 0 \quad \text{as } n \to +\infty$$

(in particular, if $\xi_n|\alpha_n| \to +\infty$), then f is differentiable at no *point of $\mathbb{R}$.*

Proof. This is the logical contrapositive of Theorem 1: the statement "f differentiable at some point $\Rightarrow \xi_n|\alpha_n| \to 0$" is equivalent to "$\xi_n|\alpha_n| \not\to 0 \Rightarrow f$ differentiable at no point." □

The continuity of f on $\mathbb{R}$ is almost evident. Indeed, the absolute convergence of the series $\sum \alpha_n$ guarantees that the series of functions defining f converges uniformly absolutely on $\mathbb{R}$ (Lemma 1), and consequently that its sum is continuous there.

2.2. *Applications and remarkable examples*

This section presents concrete examples of continuous but not differentiable functions, illustrating the generalized Weierstrass phenomenon.

Example 1 (Function with Geometric Frequencies). The function defined by

$$f_1(x) = \sum_{n=1}^{+\infty} \frac{e^{2^n i\pi x}}{n^2},$$

is continuous on $\mathbb{R}$ but nowhere differentiable. Here $\xi_n = 2^n\pi$ and $\alpha_n = 1/n^2$.

Lacunary condition: $\xi_{n+1}/\xi_n = 2$ for all n, so the condition $\xi_{n+1} \geq \eta\xi_n$ holds with $\eta = 2 > 1$.

Non-differentiability: $\xi_n|\alpha_n| = \frac{2^n\pi}{n^2} \to +\infty$, so by Corollary 1, f_1 is nowhere differentiable.

Example 2 (Function with Factorial Frequencies). The function defined by

$$f_2(x) = \sum_{n=0}^{+\infty} \frac{e^{i\,n!\,x}}{2^n},$$

constitutes another remarkable example of a continuous, nowhere differentiable function on $\mathbb{R}$. Here $\xi_n = n!$ and $\alpha_n = 1/2^n$.

Lacunary condition: $\xi_{n+1}/\xi_n = (n+1)!/n! = n+1 \geq 2$ for all $n \geq 1$, so the condition holds with $\eta = 2 > 1$.

Non-differentiability: $\xi_n|\alpha_n| = n!/2^n \to +\infty$, so by Corollary 1, f_2 is nowhere differentiable.

Example 3. The functions

$$f_3(x) = \sum_{n=1}^{+\infty} \frac{\cos(2^n\pi x)}{n^2} \quad \text{and} \quad f_4(x) = \sum_{n=0}^{+\infty} \frac{\sin(n!\,x)}{2^n},$$

are continuous on $\mathbb{R}$ but nowhere differentiable.

Verification for f_3: $\xi_n = 2^n\pi$, $\alpha_n = 1/n^2$ — identical to Example 1, so $\eta = 2$ and $\xi_n|\alpha_n| \to +\infty$.

Verification for f_4: $\xi_n = n!$, $\alpha_n = 1/2^n$ — identical to Example 2, so $\eta = 2$ and $\xi_n|\alpha_n| \to +\infty$.

Both conclusions follow from Theorem 2 (the real-valued analogue of Corollary 1).

Assume f is differentiable at 0. Then f can be written in the form

$$x \mapsto f(0) + f'(0)x + xF(x),$$

for some function F continuous on $\mathbb{R}$ (extended continuously at 0 by the definition of the derivative). We show that F is bounded on $\mathbb{R}$ in two steps:

- **On** $[-1, 1]$**:** F is continuous on the compact interval $[-1, 1]$, hence bounded there by the Extreme Value Theorem.
- **On** $\mathbb{R} \setminus (-1, 1)$**:** for such x,

$$|F(x)| = \left|\frac{f(x) - f(0) - f'(0)x}{x}\right| \leq |f(x)| + |f(0)| + |f'(0)|,$$

$$\leq \sum_{j=0}^{+\infty} |\alpha_j| + |f(0)| + |f'(0)|,$$

  which is a finite constant independent of x.

Combining both bounds, F is bounded on all of $\mathbb{R}$.

Independently, the automorphism property of the Fourier transform on $\mathcal{S}(\mathbb{R},\mathbb{C})$ allows us to construct a suitable function ψ. Explicitly, let $\widehat{\psi}$ be in $C^\infty(\mathbb{R})$ a bump function (a standard smooth Urysohn-type function, whose existence is classical, see e.g. Hörmander, *The Analysis of Linear Partial Differential Operators I*) supported in $[-\eta,-1/\eta]$ and satisfying $\widehat{\psi}(-1)=1$. Since $\widehat{\psi}\in\mathcal{S}(\mathbb{R},\mathbb{C})$ and the Fourier transform is an automorphism of $\mathcal{S}(\mathbb{R},\mathbb{C})$, its inverse Fourier transform ψ also belongs to $\mathcal{S}(\mathbb{R},\mathbb{C})$, as required. Since F is bounded on $\mathbb{R}$ (as shown above), let $C:=\sup_{\mathbb{R}}|F|<\infty$. Then for all $n\in\mathbb{N}$ and all $x\in\mathbb{R}$:

$$\left|xF\left(\frac{x}{\xi_n}\right)\psi(x)\right|\leq C|x\psi(x)|,$$

and the dominating function $x\mapsto C|x\psi(x)|$ is integrable on $\mathbb{R}$ since $\psi\in\mathcal{S}(\mathbb{R},\mathbb{C})$. Moreover, by continuity of F at 0, $F(x/\xi_n)\to F(0)$ point wise as $n\to\infty$, for every fixed x. The Dominated Convergence Theorem (see Appendix), therefore, applies and yields,

$$\lim_{n\to+\infty}\int_{\mathbb{R}}xF\left(\frac{x}{\xi_n}\right)\psi(x)\,dx=F(0)\int_{\mathbb{R}}x\psi(x)\,dx,$$

which, combined with the earlier computation $\xi_n\alpha_n=\int_{\mathbb{R}}xF(x/\xi_n)\psi(x)\,dx$, gives the desired conclusion $\lim_{n\to+\infty}\xi_n|\alpha_n|=0$.

The underlying mechanism for the non-differentiability of these functions lies in the interaction between two complementary phenomena:

- The extremely rapid growth of the frequencies ξ_n, which generates oscillations at all spatial scales.
- The controlled but non-negligible amplitude of the coefficients α_n, which preserves the cumulative effect of these oscillations.

The condition $\xi_{n+1}\geq\eta\,\xi_n$ with $\eta>1$ guarantees that each new term introduces oscillations that are significantly finer than the previous one, thereby preventing the formation of a tangent at any point.

These functions typically exhibit a fractal structure characterized by an approximate self-similarity. This structure manifests itself through several properties:

- Scale invariance: regardless of the zoom level applied, the function exhibits oscillations of comparable amplitude.
- Constant local complexity: the restriction of the function to any non-trivial interval reveals the same degree of irregularity.
- Sensitivity to initial conditions: slight variations in the argument lead to significant changes in the value of the function.

The following graphical illustrations illustrate these remarkable properties.

The examples f_1 and f_2 illustrate the diversity of possible behaviors within this class of functions:

- Function f_1, with its geometric frequencies, exhibits apparent regularity at large scales but extreme local irregularity.

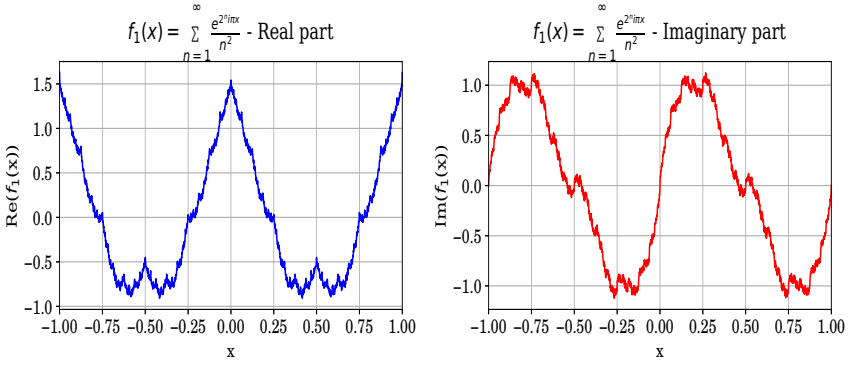


Fig. 1. Function f_1 — Real part (blue) and imaginary part (red).

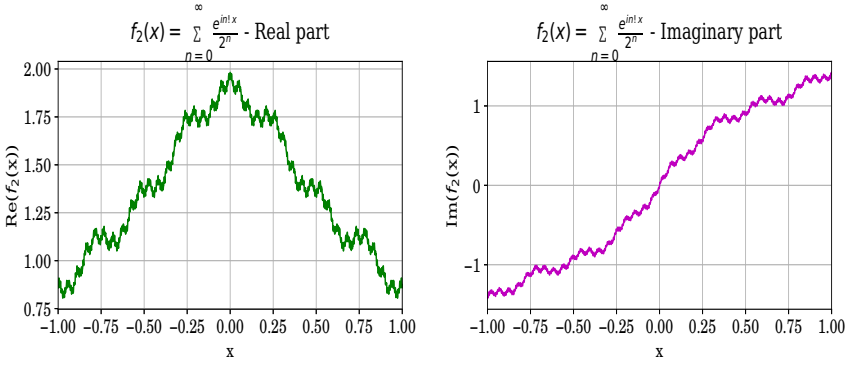


Fig. 2. Function f_2 — Real part (green) and imaginary part (orange).

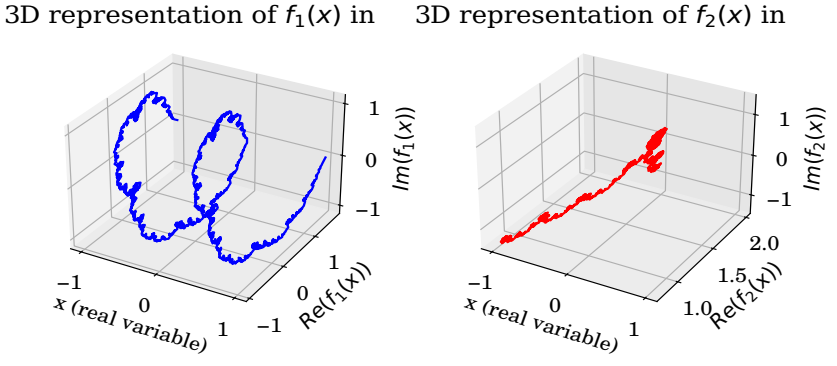


Fig. 3. 3D representations of functions f_1 and f_2 in the space $(x, \mathrm{Re}(f(x)), \mathrm{Im}(f(x)))$.

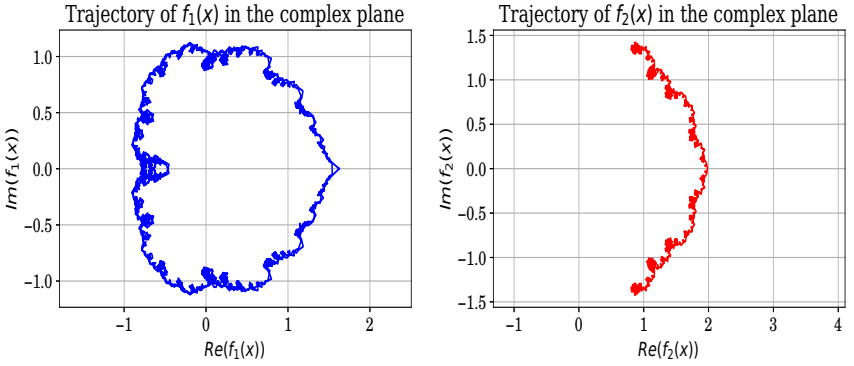


Fig. 4. Trajectories of functions f_1 and f_2 in the complex plane.

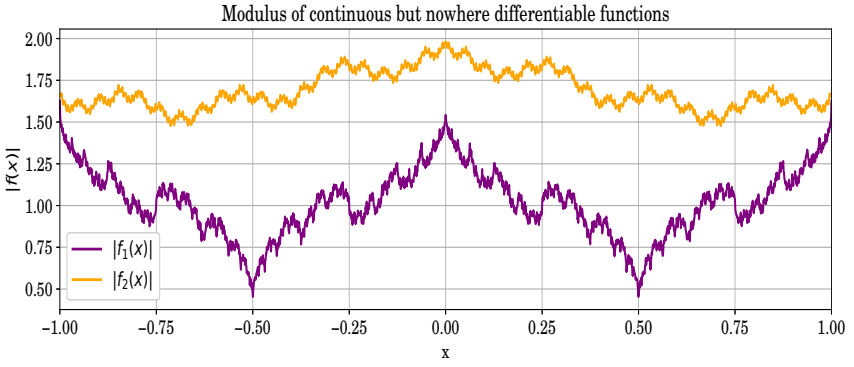


Fig. 5. Graphs of the moduli $|f_1(x)|$ and $|f_2(x)|$ as functions of x.

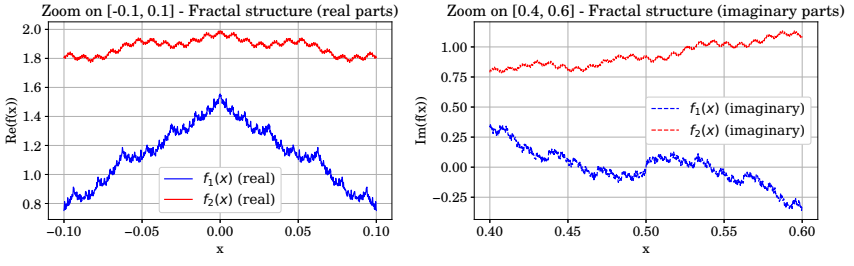


Fig. 6. Successive zooms revealing the self-similar fractal structure.

- Function f_2, combining exponential decay and factorial growth, shows more complex behavior even at intermediate scales.
- Both functions verify the generalized Weierstrass phenomenon and possess an intrinsic fractal structure.

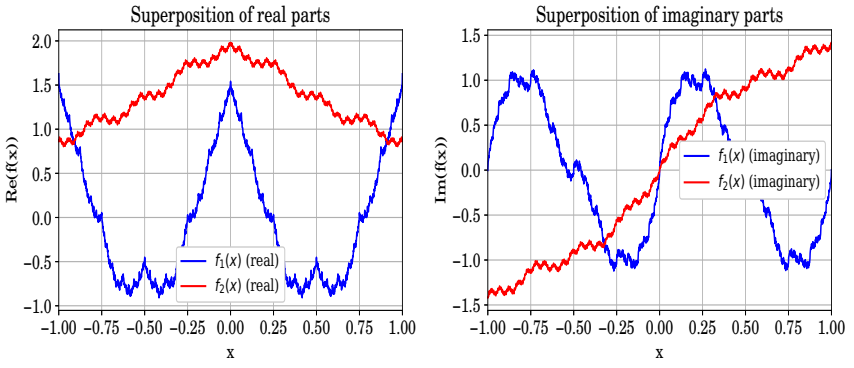


Fig. 7. Superposition of real parts of f_1 and f_2.

This study sheds light on the subtle nature of continuity and the limits of classical geometric intuition, while providing valuable tools for modern analysis of complex, multi-scale phenomena. We will need some classical tools for our work, essentially related to Fourier analysis.

2.3. *Schwartz space and Fourier Transform (Tools)*

Definition–Theorem 1 (Schwartz Space and Fourier Transform).

(1) **Schwartz Space**: The *Schwartz space on* $\mathbb{R}$, denoted $\mathcal{S}(\mathbb{R},\mathbb{C})$, is the complex vector space of functions $f \in \mathcal{C}^\infty(\mathbb{R},\mathbb{C})$ such that $\lim_{x\to\pm\infty} x^q f^{(p)}(x) = 0$ for all $q,p \in \mathbb{N}$. Consequently, $\mathcal{S}(\mathbb{R},\mathbb{C}) \subset \mathcal{L}^1(\mathbb{R},\mathbb{C})$.

(2) **Fourier Transform**: For any $f \in \mathcal{S}(\mathbb{R},\mathbb{C})$, the *Fourier transform of* f, denoted $\widehat{f}$, is the function $v \mapsto \int_{\mathbb{R}} f(x)e^{-ivx}\,dx$. The mapping $f \mapsto \widehat{f}$ is an automorphism of $\mathcal{S}(\mathbb{R},\mathbb{C})$ into itself, and for all $f \in \mathcal{S}(\mathbb{R},\mathbb{C})$ and $x \in \mathbb{R}$:

$$f(x) = \int_{\mathbb{R}} \widehat{f}(v)e^{ivx}\,dv.$$

Furthermore, for all $f \in \mathcal{S}(\mathbb{R},\mathbb{C})$ and $v \in \mathbb{R}$: $\widehat{f'} = -i\widehat{f_d}$, where f_d denotes the function $x \mapsto xf(x)$.

This is all that is required to prove the principal theorem.

2.4. *Proof of the Principal Theorem*

Problem reduction

Let $x_0 \in \mathbb{R}$. For every $x \in \mathbb{R}$: $f(x) = \sum_{j=0}^{+\infty} \beta_j e^{i\xi_j(x-x_0)} = g(x-x_0)$, where $\beta_n = \alpha_n e^{i\xi_n x_0}$ for all $n \in \mathbb{N}$ and $g(x) = \sum_{j=0}^{+\infty} \beta_j e^{i\xi_j x}$ for all $x \in \mathbb{R}$. The series $\sum \beta_n$ is absolutely convergent, and the differentiability of f at x_0 is equivalent to that of g at 0. Moreover,

$$\lim_{n\to+\infty} \xi_n|\alpha_n| = 0 \;\Leftrightarrow\; \lim_{n\to+\infty} \xi_n|\beta_n| = 0.$$

Thus, it suffices to prove the theorem for differentiability in 0 to establish it in the general case.

Proof of the Theorem at the point 0

Assume f is differentiable at 0. Then f can be written in the form

$$x \mapsto f(0) + f'(0)x + xF(x),$$

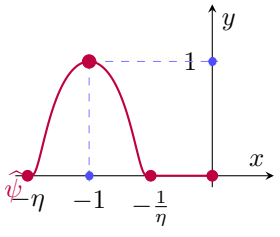
for some function $F \in \mathcal{C}(\mathbb{R}, \mathbb{C})$. For every $x \in \mathbb{R} \setminus (-1, 1)$:

$$|F(x)| = \left| \frac{f(x) - f(0) - f'(0)x}{x} \right| \leq |f(x)| + |f(0)| + |f'(0)|$$

$$\leq \sum_{j=0}^{+\infty} |\alpha_j| + |f(0)| + |f'(0)|,$$

so F is bounded on $\mathbb{R}$ by the Extreme Value Theorem.

Independently, the bijectivity of the Fourier transform on $\mathcal{S}(\mathbb{R}, \mathbb{C})$ allows us to choose a function $\psi \in \mathcal{S}(\mathbb{R}, \mathbb{C})$ such that $\widehat{\psi}$ is of class $\mathcal{C}^\infty$ on $\mathbb{R}$, has compact support contained in $[-\eta, -\frac{1}{\eta}]$, and satisfies $\widehat{\psi}(-1) = 1$.



This function will serve to *isolate* the amplitude α_n of the harmonic $e^{i\xi_n x}$ of f for every $n \in \mathbb{N}$. Indeed, for every $n \in \mathbb{N}$, the absolute convergence of the series $\sum \alpha_n$ allows us to interchange summation and integration in the following calculation:

$$\int_{\mathbb{R}} f\left(\frac{x}{\xi_n}\right) \psi(x) dx = \sum_{j=0}^{+\infty} \alpha_j \int_{\mathbb{R}} \psi(x) e^{\frac{i\xi_j x}{\xi_n}} dx = \sum_{j=0}^{+\infty} \alpha_j \widehat{\psi}\left(-\frac{\xi_j}{\xi_n}\right).$$

Now, for every $j \in [0, n-1]$, $\widehat{\psi}(-\frac{\xi_j}{\xi_n}) = 0$ because $\frac{\xi_j}{\xi_n} \leq \frac{1}{\eta}$, and for every $j > n$, $\widehat{\psi}(-\frac{\xi_j}{\xi_n}) = 0$ because $\frac{\xi_j}{\xi_n} \geq \eta$. Finally, since $\widehat{\psi}(-1) = 1$, we have $\int_{\mathbb{R}} f(\frac{x}{\xi_n})\psi(x)dx = \alpha_n$. We will now proceed to a second calculation of the same integral. For every $n \in \mathbb{N}$:

$$\begin{aligned}
\xi_n \alpha_n &= \xi_n \int_{\mathbb{R}} f\left(\frac{x}{\xi_n}\right) \psi(x) dx, \\
&= \xi_n \int_{\mathbb{R}} \left(f(0) + f'(0)\frac{x}{\xi_n} + \frac{x}{\xi_n} F\left(\frac{x}{\xi_n}\right) \right) \psi(x) dx, \\
&= \xi_n f(0) \int_{\mathbb{R}} \psi(x) dx + f'(0) \int_{\mathbb{R}} x\psi(x) dx + \int_{\mathbb{R}} xF\left(\frac{x}{\xi_n}\right) \psi(x) dx, \\
&= \xi_n f(0) \widehat{\psi}(0) + i f'(0) \widehat{\psi}'(0) + \int_{\mathbb{R}} xF\left(\frac{x}{\xi_n}\right) \psi(x) dx, \\
&= \int_{\mathbb{R}} xF\left(\frac{x}{\xi_n}\right) \psi(x) dx.
\end{aligned}$$

The function F being bounded on $\mathbb{R}$ and continuous at 0, the Dominated Convergence Theorem finally shows, as desired, that $\lim_{n\to+\infty} \xi_n|\alpha_n| = 0$.

2.5. *The case of real trigonometric series*

The work we have just carried out on complex trigonometric series readily extends to the case of real trigonometric series.

Theorem 2 (A Necessary Condition for Differentiability for Certain Real Trigonometric Series). *Let $(\alpha_n)_{n\in\mathbb{N}}$ be a real sequence, $(\xi_n)_{n\in\mathbb{N}}$ a sequence of strictly positive real numbers, and $\eta > 1$. Assume that the series $\sum \alpha_n$ converges absolutely and that $\xi_{n+1} \geq \eta\,\xi_n$ for all $n \in \mathbb{N}$. Define for every $x \in \mathbb{R}$:*

$$\mathscr{R}(x) = \sum_{j=0}^{+\infty} \alpha_j \cos(\xi_j x) \quad \textit{and} \quad \mathscr{I}(x) = \sum_{j=0}^{+\infty} \alpha_j \sin(\xi_j x).$$

The functions $\mathscr{R}$ and $\mathscr{I}$ are continuous in $\mathbb{R}$, and if they are differentiable at least one point, then $\lim_{n\to+\infty} \xi_n|\alpha_n| = 0$.

Proof. The reasoning employed in the complex case applies without change to the real-valued functions

$$x \overset{\mathscr{R}}{\longmapsto} \frac{1}{2}\left(\sum_{j=0}^{\infty} \alpha_j e^{i\xi_j x} + \sum_{j=0}^{\infty} \alpha_j e^{-i\xi_j x}\right),$$

and

$$x \overset{\mathscr{I}}{\longmapsto} \frac{1}{2i}\left(\sum_{j=0}^{\infty} \alpha_j e^{i\xi_j x} - \sum_{j=0}^{\infty} \alpha_j e^{-i\xi_j x}\right).$$

The result then follows directly from Theorem (1). □

Example 4. The functions $x \overset{f_3}{\longmapsto} \sum_{n=1}^{+\infty} \frac{\cos(2^n \pi x)}{n^2}$ and $x \overset{f_4}{\longmapsto} \sum_{n=0}^{+\infty} \frac{\sin(n!\,x)}{2^n}$ are continuous in $\mathbb{R}$ but differentiable nowhere.

3. Weierstrass Functions: Historical Examples

Weierstrass functions represent a major contribution to 19th-century analysis. In 1872, Weierstrass presented the first continuous function on $\mathbb{R}$ but differentiated at no point.[5] This surprising discovery predates the work of Bolzano (1833), which remained unpublished until 1920.

Theorem 3 (Weierstrass Function). *Let $\alpha \in (0,1)$ and $\beta > 1$ be such that $\alpha\beta \geq 1$. The function defined by*

$$W_{\alpha,\beta}(x) = \sum_{k=0}^{\infty} \alpha^k \cos(\beta^k x),$$

is uniformly continuous in $\mathbb{R}$ but admits a derivative at no point[12].

Proof. Because $0 < \alpha < 1$, the geometric series $\sum_{n=0}^{\infty} \alpha^n$ converges absolutely. Furthermore, we have the relation $\beta^{n+1} > \alpha^{-1}\beta^n$ for all integers n, and note that $\alpha^{-1} > 1$. The desired conclusion now follows immediately from Theorem (1). □

3.1. *Numerical illustrations*

Three examples satisfying the hypotheses of the theorem are:

- $W_{0.5,3}(x) = \sum_{k=0}^{\infty} 0.5^k \cos(3^k x)$, $\alpha\beta = 1.5$
- $W_{0.7,2}(x) = \sum_{k=0}^{\infty} 0.7^k \cos(2^k x)$, $\alpha\beta = 1.4$
- $W_{0.9,1.2}(x) = \sum_{k=0}^{\infty} 0.9^k \cos(1.2^k x)$, $\alpha\beta = 1.08$

The fractal structure of these functions emerges under successive zooms: oscillations persist at every resolution.[11] The absence of a derivative stems from the impossibility of defining a tangent because of fluctuations at all scales. The geometric growth of the frequencies β^k prevents any local smoothing.[10]

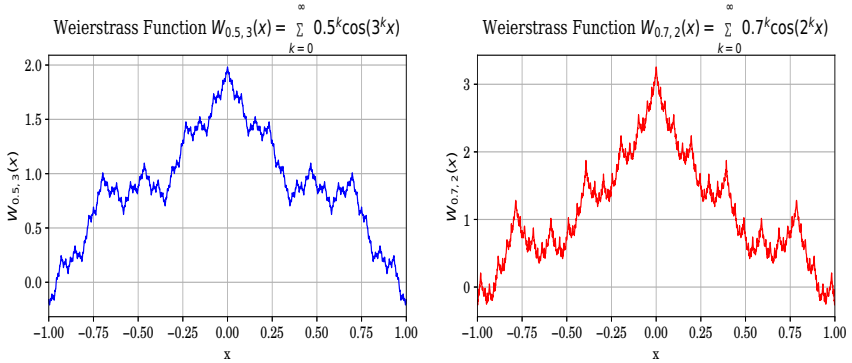


Fig. 8. Graphs of the functions $W_{0.5,3}$ and $W_{0.7,2}$ on $[-1, 1]$.

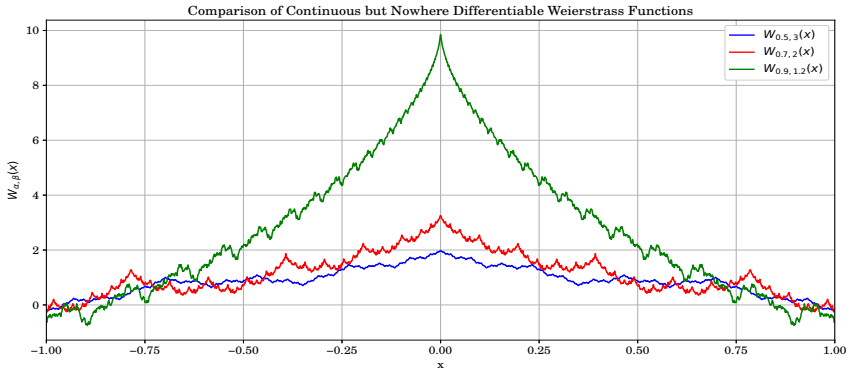


Fig. 9. Comparison of the three Weierstrass functions.

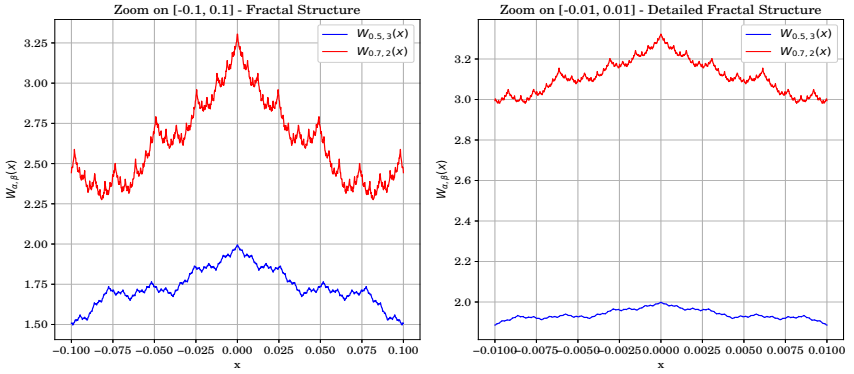


Fig. 10. Successive zooms revealing self-similarity.

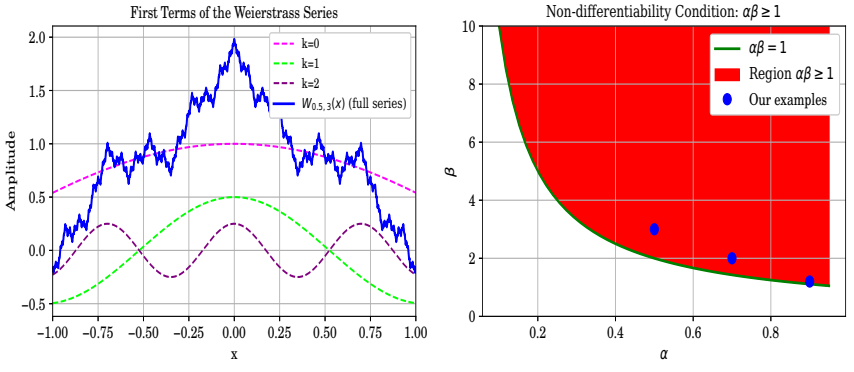


Fig. 11. Domain of parameters (α, β) satisfying $\alpha\beta \geq 1$.

3.2. *Historical impact*

This discovery profoundly influenced modern analysis:[8,9]

- ► Break with the geometric intuition associating continuity and differentiability
- ► Stimulus for the development of functional analysis
- ► Emergence of the study of pathological functions
- ► Starting point for fractal theory[11]

These functions remain essential in the teaching of advanced analysis.[13]

4. Conclusion

In this work, we established a general criterion for the continuity and nowhere differentiability of trigonometric series with decaying coefficients and rapidly increasing frequencies. The criterion clarifies the role of the interplay between coefficient decay

and frequency growth in determining differentiability and encompasses several classical constructions, including those of Riemann and Weierstrass. It also provides a systematic framework for constructing new families of continuous nowhere differentiable functions. These results contribute to a unified analytical treatment of irregularity in trigonometric series.

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Conflict of Interest

The author declares that there are no conflicts of interest related to this work.

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Appendix A

A.1. *Technical lemmas from harmonic analysis*

This appendix gathers classical analytical tools used implicitly throughout the proofs, particularly in the justification of limit processes and Fourier-analytic arguments.

Lemma A.1. *Let $(\alpha_n)_{n\in\mathbb{N}}$ be an absolutely summable sequence of complex numbers and $(\xi_n)_{n\in\mathbb{N}}$ a sequence of real numbers. Then the trigonometric series*

$$\sum_{n=0}^{\infty} \alpha_n e^{i\xi_n x},$$

converges uniformly on $\mathbb{R}$.

Proof. Since $|e^{i\xi_n x}| = 1$ for all $x \in \mathbb{R}$, the Weierstrass M-test applies directly:

$$\sum_{n=0}^{\infty} |\alpha_n e^{i\xi_n x}| = \sum_{n=0}^{\infty} |\alpha_n| < +\infty.$$

Hence, the series converges uniformly on $\mathbb{R}$. □

A.2. *Dominated convergence argument*

The proof of Theorem 1 relies on a standard application of the Dominated Convergence Theorem, which we recall here for completeness.

Lemma A.2. *Let $(F_n)_{n\in\mathbb{N}}$ be a sequence of measurable functions that point to a function F, and assume that there exists an integrable function G such that*

$$|F_n(x)| \leq G(x) \quad \text{for all } n \in \mathbb{N}, \text{ and almost every } x \in \mathbb{R}.$$

Then

$$\lim_{n\to\infty} \int_{\mathbb{R}} F_n(x)dx = \int_{\mathbb{R}} F(x)dx.$$

In our setting, the boundedness of the auxiliary function F and the rapid decay of the Schwartz function ψ ensure that the integrands are dominated by an integrable function, allowing passage to the limit in the proof of Theorem 1.

A.3. *On the isolation of frequencies*

The use of Schwartz functions with compactly supported Fourier transforms is a classical technique in harmonic analysis. It allows the isolation of a single frequency component in lacunary trigonometric series.

The condition $\xi_{n+1} \geq \eta\,\xi_n$ with $\eta > 1$ ensures that the ratios ξ_j/ξ_n are outside the support of $\widehat{\psi}$ for all $j \neq n$, which is the key mechanism that allows the extraction of the coefficient α_n through convolution.

This frequency separation principle plays a central role in many problems related to lacunary series, non-differentiability, and fractal behavior.

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